Graduate Vibe Coding Challenge

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1. Introduction

Welcome to the next generation of platform engineering. At the client's company, they are moving beyond manual code entry toward **Agentic Orchestration**. As a new graduate, your value lies not just in your ability to write syntax, but in your ability to architect systems and direct AI to execute your vision.

This challenge is a "Vibe Coding" exercise. You are the architect; the AI is the engineer. Your task is to maintain a high-level vision while ensuring technical precision through rigorous prompting.

2. Phase 1: Mindset & Culture Alignment

Our client prioritizes a "Human-in-the-Loop" mindset. Before beginning technical execution, you must establish your professional profile using the **Tagle** framework. This helps us understand your natural problem-solving "Tag" and how you enable solutions within a team.

- **Step 1: Registration.** Create your profile at https://tagle.ai/register.
- **Step 2: Discovery & Assessment.** Complete the **Initial Quiz** and the **Full Assessment** at https://tagle.ai/quiz. This is a critical component of your candidacy, be authentic and thorough.
- **Step 3: Capture Your Tag.** Navigate to your profile page and capture your "Tag" summary (e.g., Architect, Enabler, Catalyst). This must be submitted as part of your final package.

3. Phase 2: Architectural Project Selection

You must select **one** of the three projects below. Your solution must be **Python-based**, **API-first**, and utilize a **free-tier database** (e.g., SQLite, Postgres, or SQL Server Express).

Project 1: Cloud Cost Optimizer & Remediation Engine
Focus: FinOps. Build an application that ingests AWS/Azure billing exports (JSON/CSV) to identify "orphaned" resources (e.g., unattached disks, idle VMs). The tool must generate the specific CLI commands or API logic required to decommission the waste.

Project 2: Enterprise Security Guardrail Auditor
Focus: Compliance. Create a scanner that audits infrastructure configuration files (Terraform/CloudFormation) against a security baseline. It must flag high-risk patterns—such as public S3 buckets or open SSH ports—and present a visual "Risk Score" dashboard.

Project 3: Intelligent Observability & Event Watchdog
Focus: Site Reliability (SRE). Develop a service that parses application or platform logs to detect anomalies or "spikes" in errors using AI logic. When thresholds are breached, the system must trigger a simulated webhook alert and visualize health trends.

4. Phase 3: The Vibe Coding Workflow

A "Vibe Coding" workflow means you are **strictly prohibited** from writing or editing code manually. If you encounter a bug, describe the issue to your AI agent and have the agent provide the fix. You must use the same AI tool end-to-end to maintain architectural consistency.
Required Initial Execution Prompt

> **Pro-Tip: The "Staging Strategy" for Efficiency** If using a tool with a 50-request limit (like GitHub Copilot), manage your "Agent" turns wisely. Use a free LLM chat interface (like Gemini or ChatGPT) as your **Drafting Table** to pre-debug architectural concepts or refine prompts. This ensures your 50 primary Agent turns are spent on actual code assembly and deployment.

Use this exact prompt in blue text as your first message to your chosen AI agent (Claude Code, Copilot, Cursor, etc.):

"Lead Architect mode: ON. We are building a Python-based, API-first **[INSERT PROJECT CHOICE]** using a free database and a dashboard.
Rules:

- No Manual Edits: You provide all logic and fixes. I will not edit any code.
- Audit Log: You must maintain a file named prompts.md. After every turn, update that file (or provide the text block) with the prompt I just used.
- Time-Check: Start a timer. Goal is an MVP in 4-6 hours (Max window: 16h). Report 'Elapsed Time' at the end of every response. Acknowledge and let's start."

5. Final Submission and Reference

Final Submission Checklist

- [ ] Tagle.ai "Tag" output summary included.
- [ ] Public GitHub Repository link containing all source code.
- [ ] **prompts.md** file containing the full audit log of all instructions.
- [ ] AI-generated Presentation Deck (PPT or Markdown solution).
- [ ] **Confirmation:** All cloud resources decommissioned and accounts closed.

Quick Reference & Resource Links

- **Mindset (Tagle):** https://tagle.ai/register
- **GitHub Copilot:** https://github.com/features/copilot
- **AWS Free Tier:** https://aws.amazon.com/free/
- **Azure Free Tier:** https://azure.microsoft.com/free/

PROFESSIONAL RESPONSIBILITY: You are responsible for all cloud costs. Delete all resources and decommission your environment immediately after submission.